Corrected q-Binomial Inverse for h_m and Q_n

Notes (February 2026)

1 The corrected inverse formula

The forward q-binomial relation between Q_n and h_m is:

$$Q_n = \sum_{j=0}^{n} (-1)^j \, q^{\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j}. \tag{1}$$

Proposition 1. *The inverse of* (**??**) *is:*

$$h_m = \sum_{k=0}^{m} q^k \begin{bmatrix} m \\ k \end{bmatrix} Q_{m-k}. \tag{2}$$

Proof. From the E–H factorisation $\mathrm{GK}_c(z,q) = H(z) \cdot F_c(z,q)$ where $H(z) = 1/(zq;q)_\infty$ and $F_c(z,q) = (zq;q)_\infty \cdot \mathrm{GK}_c(z,q)$, we have:

$$[z^m]\, H(z) = \frac{q^m}{(q;q)_m},$$
$$[z^m]\, F_c(z,q) = \frac{Q_m}{(q;q)_m}.$$

The first identity is Euler's partition identity $1/(z;q)_\infty = \sum_{m\geq 0} z^m/(q;q)_m$ evaluated at $z \mapsto zq$, giving $[z^m] = q^m/(q;q)_m$.

Since $\mathrm{GK}_c = H \cdot F_c$, extracting $[z^m]$ gives:

$$g_m = \sum_{k=0}^{m} [z^k]\, H(z) \cdot [z^{m-k}]\, F_c(z,q) = \sum_{k=0}^{m} \frac{q^k}{(q;q)_k} \cdot \frac{Q_{m-k}}{(q;q)_{m-k}}.$$

Multiplying both sides by $(q;q)_m$ and using $h_m = (q;q)_m \cdot g_m$:

$$h_m = \sum_{k=0}^{m} \frac{(q;q)_m}{(q;q)_k \, (q;q)_{m-k}} \cdot q^k \cdot Q_{m-k} = \sum_{k=0}^{m} q^k \begin{bmatrix} m \\ k \end{bmatrix} Q_{m-k}. \qquad \square$$

Remark 2. *An earlier version of* `definitions.tex` *had* q^{j^2} *in place of* q^j *in the inverse formula. The error likely arose from confusion with the forward formula's* $q^{\binom{j+1}{2}}$*. The key difference:*

- *Forward: the sign-reversing Euler factor* $(zq;q)_\infty = \sum (-1)^j q^{\binom{j+1}{2}} z^j/(q;q)_j$ *contributes* $q^{\binom{j+1}{2}}$*.*
- *Inverse: the partition-counting factor* $1/(zq;q)_\infty = \sum q^j z^j/(q;q)_j$ *contributes* q^j*.*

2 Immediate corollary: $Q \geq 0$ implies $h \geq 0$

Corollary 3. *If* $Q_n \in \mathbb{Z}_{\geq 0}[q]$ *for all* $n \geq 0$*, then* $h_m \in \mathbb{Z}_{\geq 0}[q]$ *for all* $m \geq 0$*.*

Proof. Every term in (**??**) is a product of non-negative polynomials: $q^k \geq 0$, $\begin{bmatrix} m \\ k \end{bmatrix} \in \mathbb{Z}_{\geq 0}[q]$, and $Q_{m-k} \in \mathbb{Z}_{\geq 0}[q]$ by hypothesis. □

The converse fails: $h_m \geq 0$ does not imply $Q_n \geq 0$, because the forward formula (**??**) is an alternating sum.

3 Corrected examples for $d = 2$, $c = (1, 1, 0)$

With $Q_n = q^{n^2}$ (first Rogers–Ramanujan), the inverse gives:

$$h_m = \sum_{k=0}^{m} q^{k+(m-k)^2} \begin{bmatrix} m \\ k \end{bmatrix}.$$

$h_0 = 1$,

$h_1 = q^{0+1}\begin{bmatrix} 1 \\ 0 \end{bmatrix} + q^{1+0}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = q + q = 2q$,

$h_2 = q^{0+4}\begin{bmatrix} 2 \\ 0 \end{bmatrix} + q^{1+1}\begin{bmatrix} 2 \\ 1 \end{bmatrix} + q^{2+0}\begin{bmatrix} 2 \\ 2 \end{bmatrix} = q^4 + q^2(1+q) + q^2 = 2q^2 + q^3 + q^4$.

4 Dominance investigation (negative result)

We explored whether $Q_n \geq 0$ could be explained by coefficient-wise dominance relations between successive h_m. Define:

$$D_m = h_m - q \cdot h_{m-1},$$
$$R_n = \sum_{j=1}^{n} (-1)^{j-1} q^{\binom{j+1}{2}} \begin{bmatrix} n \\ j \end{bmatrix} h_{n-j},$$

so that $Q_n = h_n - R_n$. Computations for $d = 1, 2, 4, 5$ confirm:

1. $D_m \geq 0$ for all m and all tested profiles.
2. $R_n \geq 0$ for all n and all tested profiles.
3. Even partial sums $P_n^{(2k)} = \sum_{j=0}^{2k}(\cdots) \geq 0$.

However, all three properties are **consequences** of $Q_n \geq 0$, not independent facts:

Proposition 4. $R_n = \sum_{k=1}^{n} q^k \begin{bmatrix} n \\ k \end{bmatrix} Q_{n-k}$.

Proof. This is the inverse formula (**??**) with the $k = 0$ term (Q_n) removed: $R_n = h_n - Q_n = \sum_{k=1}^{n} q^k \begin{bmatrix} n \\ k \end{bmatrix} Q_{n-k}$. □

Proposition 5. $D_m = Q_m + \sum_{j=1}^{m} q^{2j} \begin{bmatrix} m-1 \\ j \end{bmatrix} Q_{m-j}$.

Proof. From (**??**): $D_m = \sum_{k=0}^{m} q^k \begin{bmatrix} m \\ k \end{bmatrix} Q_{m-k} - \sum_{k=0}^{m-1} q^{k+1} \begin{bmatrix} m-1 \\ k \end{bmatrix} Q_{m-1-k}$. Reindex the second sum ($k \mapsto j-1$) and use $\begin{bmatrix} m \\ j \end{bmatrix} - \begin{bmatrix} m-1 \\ j-1 \end{bmatrix} = q^j \begin{bmatrix} m-1 \\ j \end{bmatrix}$. □

Thus $D_m \geq 0$ and $R_n \geq 0$ follow from $Q_k \geq 0$ for $k \leq m$ (resp. $k \leq n-1$). The positivity of Q_n cannot be decomposed into simpler pairwise dominance inequalities via this approach.

Conclusion. The conjecture $Q_n \geq 0$ appears to be irreducibly about the interaction between the full sequence $\{h_m\}$ (encoding the CPP structure via $h_m = (q;q)_m \cdot g_m$) and the q-binomial transform. Proving it likely requires a direct construction (bijective or algebraic) rather than reduction to simpler inequalities.